

JavaScript Object Management

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Adding a New Property

You can add new properties to an existing object using `Object.defineProperty()` :

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Object Management</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Adding a New Property</h4>
<p id="demo"></p>

<script>
const person = {
  firstName: "John",
  lastName: "Doe"
};

// Add a new property
Object.defineProperty(person, "fullName", {
  value: "John Doe",
  writable: true
});

document.getElementById("demo").innerHTML = person.fullName;
</script>

</body>
</html>
```

Example 1

Result [View Example](#)

Changing a Property Value

You can change a property value using `Object.defineProperty()` :

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Object Management</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Changing a Property Value</h4>
<p id="demo"></p>

<script>
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 30
};

// Change the age property
Object.defineProperty(person, "age", {
  value: 35
});

document.getElementById("demo").innerHTML = "Age: " + person.age;
</script>

</body>
</html>
```

Example 2

Result [View Example](#)

Property Attributes

Property attributes control how properties behave:

- **writable** - If `true`, the value can be changed
- **enumerable** - If `true`, the property shows up in enumeration (`for...in`)
- **configurable** - If `true`, the property can be deleted or changed

List All Object Properties

`Object.getOwnPropertyNames()` returns all properties (including non-enumerable):

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Object Management</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>List All Object Properties</h4>
<p id="demo"></p>

<script>
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 30,
  eyeColor: "blue"
};

// Get all property names
const properties = Object.getOwnPropertyNames(person);
let text = "";
for (let prop of properties) {
  text += prop + ": " + person[prop] + "<br>";
}
document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>
```

Example 3

Result [View Example](#)

Adding Getters and Setters

`Object.defineProperty()` can also add getters and setters:

```

<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Object Management</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Adding Getters and Setters</h4>
<p id="demo"></p>

<script>
const person = {
  firstName: "John",
  lastName: "Doe"
};

Object.defineProperty(person, "fullName", {
  get: function() {
    return this.firstName + " " + this.lastName;
  },
  set: function(value) {
    const parts = value.split(" ");
    this.firstName = parts[0];
    this.lastName = parts[1];
  },
  enumerable: true
});

document.getElementById("demo").innerHTML = "Before: " + person.fullName + "<br>";
person.fullName = "Jane Smith";
document.getElementById("demo").innerHTML += "After: " + person.fullName;
</script>

</body>
</html>

```

Example 4

Result [View Example](#)

A Counter Example

Using `Object.defineProperty()` to create a counter with controlled access:

```

<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Object Management</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>A Counter Example</h4>
<p id="demo"></p>

<script>
const obj = { counter: 0 };

Object.defineProperty(obj, "increment", {
  get: function() {
    this.counter++;
    return this.counter;
  }
});

Object.defineProperty(obj, "decrement", {
  get: function() {
    this.counter--;
    return this.counter;
  }
});

Object.defineProperty(obj, "reset", {
  get: function() {
    this.counter = 0;
    return this.counter;
  }
});

document.getElementById("demo").innerHTML =
  "Initial: " + obj.reset + "<br>" +
  "After increment: " + obj.increment + "<br>" +
  "After increment: " + obj.increment + "<br>" +
  "After decrement: " + obj.decrement + "<br>" +
  "After reset: " + obj.reset;
</script>

</body>
</html>

```

Example 5

Result [View Example](#)

Document

Document in project

You can [Download PDF](#) file.

Reference

- [W3Schools JavaScript Object Management](#)